Circles

Test 1

Date: _____ Duration: 1 hour 30 minutes Maximum marks: 40

All questions are compulsory. Draw a rough diagram wherever it helps. Show all your steps and give reasons.

Paper design. The paper follows the CBSE pattern. Section A: 6×1 mark (4 multiple-choice questions, 1 fill-in-the-blank question and 1 assertion–reason question). Section B: 4×2 marks. Section C: 4×3 marks. Section D: 2×5 marks. Section E: 1 case-based question of 4 marks. Total: 40 marks.

Theorem numbers. The paper uses the theorem numbers of the class notes (Circles classes 1–4). Theorems 1–4 carry about 27 marks. Theorems 5–8 carry about 13 marks. Theorems 9–12 are not assessed.

Competency focus. Question 2 tests direct use of a result. The other questions are competency-based. Students find an error (Q7), work backwards (Q10, Q16), write proofs (Q12, Q15), construct a figure (Q13), compare chords (Q5, Q14) and apply results to real situations (Q8, Q13, Q16, Q17).

Concepts assessed. Equal chords and angles at the centre, Theorems 1 and 2 (Q1, 8, 12, 17); the line from the centre to a chord, Theorems 3 and 4 (Q2, 3, 6, 7, 9, 10, 11, 14, 15, 16, 17); the circle through three points, Theorem 5 (Q4, 13); chords and their distance from the centre, Theorems 6–8 (Q5, 9, 14, 16, 17).

Common misconceptions. Joining the centre to any point of a chord and taking the line as perpendicular (Q7); using the full chord in place of half the chord in Pythagoras theorem (Q2, 10, 11); adding the distances in every case of parallel chords (Q11).

Marking. Award method marks for a correct set-up, even where the final computation contains an arithmetic error. In proofs, award marks for the correct congruence rule with all three conditions and their reasons.

Section A (1 mark each)

Questions 1, 2, 3 and 5 are multiple choice. Question 4 is fill in the blank. Question 6 is assertion–reason.

1. In a circle with centre O, chords AB and CD are equal and $\angle OAB = 55^\circ$. What is $\angle COD$? [1]

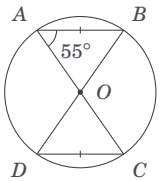
(a) 55° (b) 70°

(c) 110° (d) 125°

Solution.

Answer: (b) 70°.

1. $OA = OB$ (radii), so $\angle OBA = \angle OAB = 55^\circ$.
2. $\angle AOB = 180^\circ - 55^\circ - 55^\circ = 70^\circ$ (angle sum of a triangle).
3. $\angle COD = \angle AOB = 70^\circ$ (equal chords subtend equal angles at the centre, Theorem 1).



2. A chord of length 20 cm is drawn in a circle of radius 26 cm. What is the distance of the chord from the centre? [1]

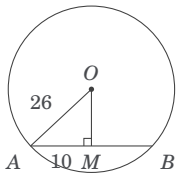
(a) 6 cm (b) 16 cm

(c) 24 cm (d) 28 cm

Solution.

Answer: (c) 24 cm.

1. $OM \perp AB$, so $AM = 10$ cm (the perpendicular from the centre bisects the chord, Theorem 4).
2. $OM^2 = 26^2 - 10^2 = 676 - 100 = 576$, so $OM = 24$ cm (Pythagoras theorem).



3. Which statement about a circle with centre O is **false**? [1]

(a) The line joining O to the midpoint of a chord (not a diameter) is perpendicular to the chord.

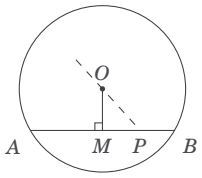
(b) The perpendicular from O to a chord bisects the chord.

(c) The perpendicular bisector of every chord passes through O.

(d) Every line through O bisects every chord of the circle.

Solution.

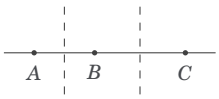
Answer: (d). A line through O that is slanted to a chord (not a diameter) meets the chord away from its midpoint. Statements (a), (b) and (c) are Theorems 3 and 4 and their consequence.



4. Fill in the blank: The number of circles that pass through three points lying on one straight line is _____. [1]

Solution.

Answer: 0 (none). The perpendicular bisectors of AB and BC are parallel, so they never meet. No point is at the same distance from all three points. (Theorem 5 needs three non-collinear points.)



5. A circle has radius 10 cm. It has two chords of lengths 12 cm and 18 cm. Which chord is nearer the centre? [1]

(a) The 12 cm chord

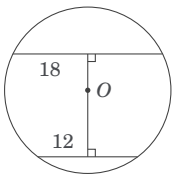
(b) The 18 cm chord

(c) Both are at the same distance

(d) It cannot be decided without more information

Solution.

Answer: (b). Of two chords of a circle, the longer chord is nearer the centre (Theorem 8).



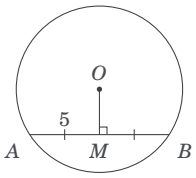
6. Assertion–reason. *Choose: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true, but R is false. (d) A is false, but R is true.*

Assertion (A): In a circle with centre O, $OM \perp AB$, where AB is a chord and M lies on AB. If $AM = 5$ cm, then $AB = 10$ cm.

Reason (R): The perpendicular from the centre of a circle to a chord bisects the chord. [1]

Solution.

Answer: (a). $OM \perp AB$, so M is the midpoint of AB (Theorem 4). So $AB = 2 \times 5 = 10$ cm. R is the reason for A.



Section B (2 marks each)

7. In a circle with centre O, chord $AB = 10$ cm. P is a point on AB with $AP = 4$ cm. Meena writes: "OP is perpendicular to AB, because the line joining the centre to a point of a chord is perpendicular to the chord."

(a) What is wrong in Meena's reasoning? [1]

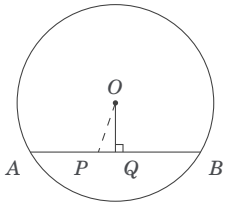
(b) Which point Q of AB must be joined to O so that $OQ \perp AB$? How far is Q from A? [1]

Solution.

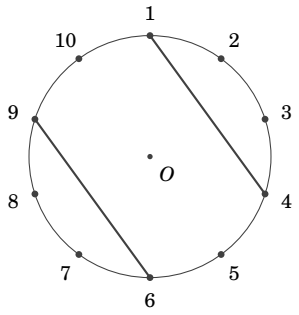
(a) Theorem 3 is true only for the line from the centre to the *midpoint* of the chord. P is not the midpoint: $AP = 4$ cm but $PB = 6$ cm. So OP is not perpendicular to AB.

(b) Q is the midpoint of AB (Theorem 3), so $AQ = \dfrac{10}{2} = 5$ cm.

Answer: Q is the midpoint of AB; $AQ = 5$ cm.



8. Divya draws a circle with centre O for a rangoli. She marks 10 dots on the circle at equal distances and numbers them 1 to 10 in order. She joins dot 1 to dot 4, and dot 6 to dot 9 with straight lines.



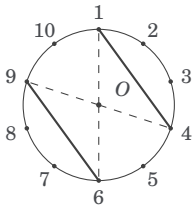
(a) Find the angle that the line from dot 1 to dot 4 subtends at O. [1]

(b) Show that the two lines are equal in length. [1]

Solution.

Given: 10 equally spaced dots on a circle with centre O; chords from dot 1 to dot 4 and from dot 6 to dot 9.

To find: (a) the angle at O; (b) show the chords are equal.



1. The 10 dots divide the full angle 360° at O into 10 equal angles of $\dfrac{360°}{10} = 36°$.
2. From dot 1 to dot 4 there are 3 gaps, so the angle is $3 \times 36° = 108°$.
3. From dot 6 to dot 9 there are also 3 gaps, so that angle is also 108°.
4. The two chords subtend equal angles at the centre, so they are equal (Theorem 2).

Answer: (a) 108°; (b) the chords are equal.

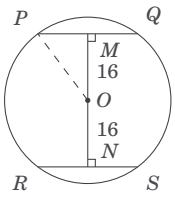
9. Chords PQ and RS of a circle with centre O are each 16 cm from O, and $PQ = 24$ cm.

(a) Find RS. Give the reason. [1]

(b) Find the radius of the circle. [1]

Solution.

Given: $OM \perp PQ$, $ON \perp RS$, $OM = ON = 16$ cm, $PQ = 24$ cm.
To find: RS and the radius.



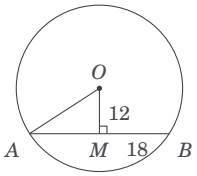
1. PQ and RS are equidistant from O, so $RS = PQ = 24$ cm (Theorem 7).
2. $PM = \dfrac{24}{2} = 12$ cm (Theorem 4).
3. $OP^2 = OM^2 + PM^2 = 256 + 144 = 400$, so $OP = 20$ cm (Pythagoras theorem).

Answer: $RS = 24$ cm; radius 20 cm.

10. A chord of a circle is 18 cm long and is 12 cm from the centre. Find the diameter of the circle. Draw a rough diagram that shows the circle, its centre O, the chord AB and the perpendicular OM from O to AB. Mark the given lengths on it. [2]

Solution.

Given: chord $AB = 18$ cm, $OM \perp AB$, $OM = 12$ cm.
To find: the diameter.



1. $AM = \dfrac{18}{2} = 9$ cm (Theorem 4).
2. $OA^2 = OM^2 + AM^2 = 144 + 81 = 225$, so $OA = 15$ cm (Pythagoras theorem).
3. Diameter $= 2 \times 15 = 30$ cm.

Answer: 30 cm.

Section C (3 marks each)

11. AB and CD are parallel chords of a circle with centre O and radius 25 cm. $AB = 40$ cm and $CD = 30$ cm.

(a) Find the distance of each chord from O. [1]

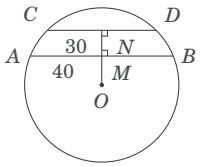
(b) Find the distance between the chords when they lie on the same side of O. [1]

(c) Find the distance between the chords when they lie on opposite sides of O. [1]

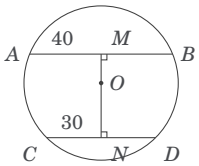
For (b) and (c), draw a rough diagram of each case. Show the circle, its centre O, both chords, and one line through O perpendicular to both chords.

Solution.

Given: radius 25 cm, $AB \parallel CD$, $AB = 40$ cm, $CD = 30$ cm; $OM \perp AB$, $ON \perp CD$. **To find:** OM, ON and the distance MN in both cases.



same side

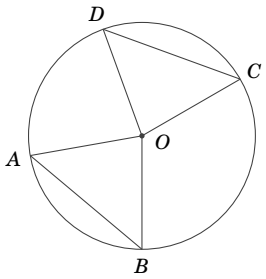


opposite sides

1. $AM = 20$ cm and $CN = 15$ cm (Theorem 4).
2. $OM^2 = 25^2 - 20^2 = 625 - 400 = 225$, so $OM = 15$ cm (Pythagoras theorem).
3. $ON^2 = 25^2 - 15^2 = 625 - 225 = 400$, so $ON = 20$ cm.
4. M, O, N lie on one line perpendicular to both chords. Same side: $MN = ON - OM = 20 - 15 = 5$ cm.
5. Opposite sides: $MN = OM + ON = 15 + 20 = 35$ cm.

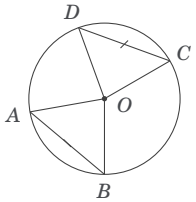
Answer: (a) AB is 15 cm and CD is 20 cm from O; (b) 5 cm; (c) 35 cm.

12. AB and CD are equal chords of a circle with centre O. Prove that $\angle OAB = \angle OCD$. [3]



Solution.

Given: AB and CD are chords of a circle with centre O, $AB = CD$. **To prove:** $\angle OAB = \angle OCD$. **Construction:** join OA, OB, OC, OD. Proof: In $\triangle OAB$ and $\triangle OCD$:



1. $OA = OC$ (radii).
2. $OB = OD$ (radii).
3. $AB = CD$ (given).
4. So $\triangle OAB \cong \triangle OCD$ (SSS rule).
5. So $\angle OAB = \angle OCD$ (corresponding parts of congruent triangles).

Hence proved.

13. Three houses A, B and C stand in a village. $AB = 300$ m, $BC = 400$ m and $\angle ABC = 90°$. The panchayat wants to dig a well at a point W that is at the same distance from all three houses.

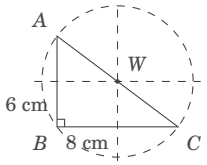
(a) Use the scale 1 cm = 50 m. Draw $AB = 6$ cm. At B, draw $\angle ABC = 90°$ and mark C with $BC = 8$ cm. Join AC. [1]

(b) Construct the perpendicular bisectors of AB and BC. Mark their meeting point W. [1]

(c) Measure WA. Find the real distance of the well from each house. [1]

Solution.

Given: $AB = 300$ m, $BC = 400$ m, $\angle B = 90°$; scale 1 cm = 50 m. **To find:** the position of W and its distance from each house.



1. $AB = \dfrac{300}{50} = 6$ cm and $BC = \dfrac{400}{50} = 8$ cm on paper.
2. Every point on the perpendicular bisector of AB is equally far from A and B; every point on the perpendicular bisector of BC is equally far from B and C. So W is equally far from A, B and C. It is the centre of the only circle through A, B and C (Theorem 5).
3. W falls on AC, at its midpoint. $AC^2 = 6^2 + 8^2 = 100$, so $AC = 10$ cm and $WA = 5$ cm (check by measuring).
4. Real distance $= 5 \times 50 = 250$ m.

Answer: the well is 250 m from each house.

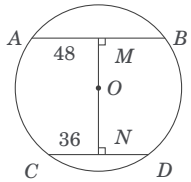
14. A circle has centre O and radius 30 cm. Chord $AB = 48$ cm and chord $CD = 36$ cm.

(a) Without calculating, say which chord is nearer the centre. Give the reason. [1]

(b) Find the distance of each chord from O, and check your answer to (a). [2]

Solution.

Given: radius 30 cm, $AB = 48$ cm, $CD = 36$ cm; $OM \perp AB$, $ON \perp CD$.
To find: which chord is nearer O; OM and ON.



1. $AB > CD$, so AB is nearer the centre (Theorem 8).
2. $AM = 24$ cm (Theorem 4); $OM^2 = 900 - 576 = 324$, so $OM = 18$ cm.
3. $CN = 18$ cm (Theorem 4); $ON^2 = 900 - 324 = 576$, so $ON = 24$ cm.
4. $18 < 24$, so AB is nearer. This agrees with (a).

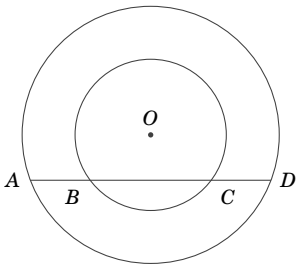
Answer: AB is nearer; $OM = 18$ cm, $ON = 24$ cm.

Section D (5 marks each)

15.

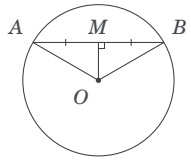
(a) Prove that the perpendicular from the centre of a circle to a chord bisects the chord. [3]

(b) Two circles have the same centre O. A straight line meets the bigger circle at A and D, and the smaller circle at B and C, in the order A, B, C, D (see the figure). Prove that $AB = CD$. [2]



Solution.

(a)

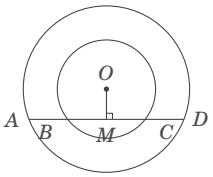


Given: a circle with centre O, chord AB, $OM \perp AB$. **To prove:** $AM = MB$. **Construction:** join OA and OB. Proof: In $\triangle OMA$ and $\triangle OMB$:

1. $\angle OMA = \angle OMB = 90°$ (given).
2. $OA = OB$ (radii; hypotenuses).
3. $OM = OM$ (common side).
4. So $\triangle OMA \cong \triangle OMB$ (RHS rule).
5. So $AM = MB$ (corresponding parts of congruent triangles).

Hence proved.

(b)

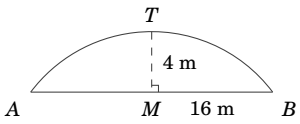


Given: two circles with centre O; a line meets them at A, B, C, D in this order. **To prove:** $AB = CD$. **Construction:** draw $OM \perp AD$. Proof:

1. AD is a chord of the bigger circle, so $AM = MD$ (Theorem 4).
2. BC is a chord of the smaller circle, so $BM = MC$ (Theorem 4).
3. $AB = AM - BM$ and $CD = MD - MC$.
4. So $AB = CD$ (equals subtracted from equals).

Hence proved.

16. The front of a temple gate has an arch. The arch is part of a circle with centre O. The ends A and B of the arch are 16 m apart. The highest point T of the arch is directly above the midpoint M of AB, and $TM = 4$ m.

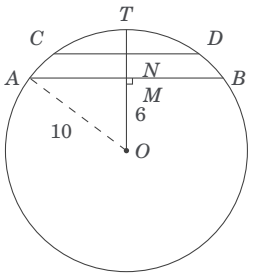


(a) Explain why the centre O lies on the line TM. [1]

(b) Let the radius be r m. Draw a rough diagram that shows the full circle, O, A, B, M and T. Find r. [2]

(c) A straight iron bar CD is fixed across the arch, parallel to AB and 2 m above it, with C and D on the arch. Find the length of the bar. Is your answer in agreement with Theorem 8? Explain. [2]

Solution.



Given: arc through A, T, B; $AB = 16$ m; M midpoint of AB; $TM \perp AB$; $TM = 4$ m. **To find:** (a) why O is on TM; (b) radius r; (c) length of CD.

(a) M is the midpoint of chord AB, so $OM \perp AB$ (Theorem 3). TM is also perpendicular to AB at M. Only one line through M is perpendicular to AB, so O lies on line TM.

(b)

1. $OT = OA = r$ (radii), so $OM = r - 4$.
2. $AM = 8$ m.
3. In right $\triangle OMA$: $r^2 = (r-4)^2 + 8^2 = r^2 - 8r + 16 + 64$.
4. So $8r = 80$ and $r = 10$ m. $OM = 6$ m.

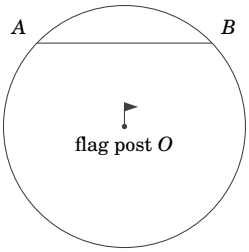
(c)

1. The bar is 2 m above AB, so its distance from O is $6 + 2 = 8$ m; let N be its midpoint, $ON \perp CD$.
2. $CN^2 = 10^2 - 8^2 = 36$, so $CN = 6$ m (Pythagoras theorem); $CD = 2 \times 6 = 12$ m (Theorem 4).
3. The bar (12 m) is shorter than AB (16 m) and farther from O (8 m against 6 m). This agrees with Theorem 8: the longer chord is nearer the centre.

Answer: (b) 10 m; (c) 12 m; yes.

Section E (case study, 4 marks)

17. Case study. The sports ground of a school is a circle with centre O and radius 29 m. A flag post stands at O. For sports day, the PE teacher draws straight chalk lines from edge to edge of the ground as running tracks. Track AB is 42 m long.



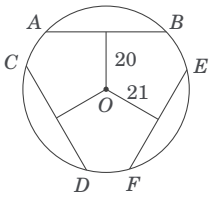
(i) Find the distance of track AB from the flag post. [1]

(ii) Track CD is also 42 m long. Without calculating, write the distance of CD from the flag post. Name the theorem you used. [1]

(iii) Compare $\angle AOB$ and $\angle COD$. Give the reason. [1]

(iv) Track EF is 40 m long. Is EF nearer to the flag post or farther from it than AB? Find its distance from the flag post. [1]

Solution.



Given: radius 29 m; $AB = CD = 42$ m; $EF = 40$ m.
(i) $AM = 21$ m (Theorem 4); $OM^2 = 29^2 - 21^2 = 841 - 441 = 400$, so $OM = 20$ m.
(ii) 20 m: equal chords are equidistant from the centre (Theorem 6).
(iii) $\angle AOB = \angle COD$: equal chords subtend equal angles at the centre (Theorem 1).
(iv) $EF < AB$, so EF is farther (Theorem 8). Half of EF is 20 m; $d^2 = 841 - 400 = 441$, so $d = 21$ m.
Answer: (i) 20 m; (ii) 20 m; (iii) equal; (iv) farther, 21 m.